PS 200C: Causal inference for social science

Causal Graphs and Causal Structural Models

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Resources for this material

Today we will just get our feet wet with a superficial exploration of the graphical approach. We don't want to get caught up in details.

But, for more DAG material:

- *What if* (Hernan and Robins), Chapters 6, 7.
- *Primer* (Pearl, Glymour, Jewell), Chapters 1.4, 1.5, 2, 3.
- Pearl, J. (2009). *Causality*
- Some online materials:
  - *Causal Diagrams: Draw Your Assumptions Before Your Conclusions* course by Miguel Hernan on EdX
  - *A Crash Course in Causality: Inferring Causal Effects from Observational Data* course by Jason Roy on Coursera.
  - Richard McElreath lecture on "Science before statistics: causal inference". `https://www.youtube.com/watch?v=KNPYUVmY3NM&t=8565s`

Goals for this lecture

- Introduction to structural causal models
- Graph basics:
  - conditional independence and d-separation
  - backdoor criterion
- Use cases for DAGs, connection to CI/SOO, handling complaints
- Important things revealed by DAGs

One point of view:

We need a language that allows us to (i) express our causal assumptions, (ii) make causal queries, (iii) determine if those queries can be answered; (iv) tell us how to get those answers.

1. To do so we can use structural causal models and their graphical representations as directed acyclic graphs
2. Potential outcomes/counterfactuals are hypothetical realizations from these structure, telling us outcomes under particular interventions.
3. Tensions between approaches once raged but have mostly subsided; in (my view of) reality these traditions are all of a piece and have different strengths and weaknesses.

Graphs and Stuctural Causal Models

Structural Causal Models (SCMs) are formal ways of specifying a set of causal relationships. They consist of:

- a group of observed variables
- a group of unobserved exogenous variables, $\{U\}$
- functions that relate them, $f_{(\cdot)}$
- an assumed distribution over the exogenous variables, $p(U)$.

E.g.,

$$
\begin{aligned}
X &= f_x(U_x) \\
A &= f_a(X, U_a) \\
Y &= f_y(X, A, U_y) \\
U_y &\perp\!\!\!\perp U_x, \;\; U_y \perp\!\!\!\perp U_a
\end{aligned}
$$

where "exogenous" ancestors, $U = \{U_x, U_y, U_z\}$ with $U \sim p(u)$.

Life in an SCM

Note that:

- $\{f_x, f_a, f_y\}$ are (but do not have to be) non-parametric functions
- they are "invariant" or "modular", don't depend on how you got there
- causal meaning to each expression, not just algebraic.

SCMs impose a deterministic, "clockwork" notion of causation:

- Population: The distribution $p(U)$ implies a distribution $p(observables)$.
- Individuals: If you knew all the values of U for a given unit and the details of every f, you would know the exact values of every variable.
- The invariance notion allow us write down what would happen (to the distributions or specific realizations) under counterfactuals that modify the model or set values in it.

Defining Causality

In this context causality means (*Primer*):

> *A variable X is a cause of a variable Y if Y in any way relies on X for its value... think of causation as a form of listening; X is a cause of Y if Y listens to X and decides its value in response to what it hears.*

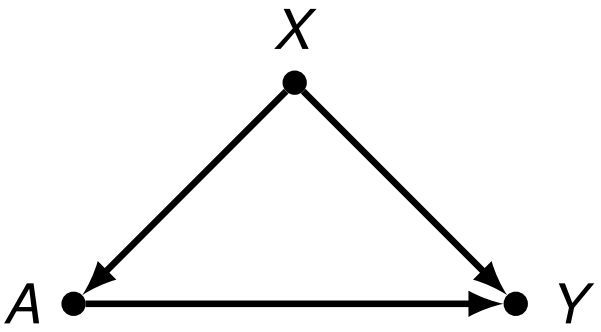
Still, specific causal estimands generally invoke potential outcomes, which are effectively tools to summarize causal relationships in this view.

SCM to Graph

For a model M such as:

$$X = f_x(U_x)$$
$$A = f_a(X, U_a)$$
$$Y = f_y(X, A, U_y)$$
$$U_y \perp\!\!\!\perp U_x, \;\; U_y \perp\!\!\!\perp U_a$$

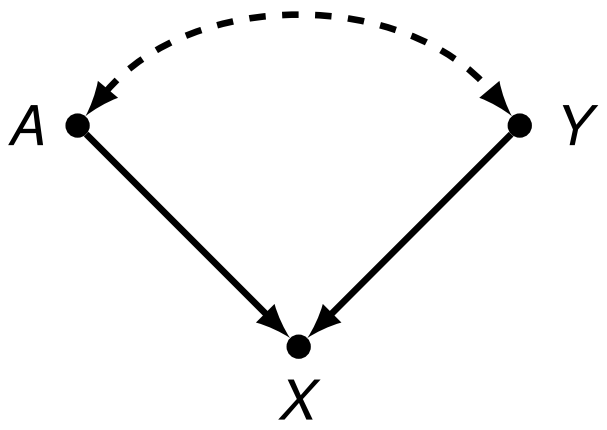
It is convenient to draw a graph, G:



- Arrows go into a node only if the corresponding argument enters the corresponding function in the SCM.
- You could show U_j pointing to each j, but usually omit these.

Missing edges, bidirected arcs

A missing edge means the value of A is not an argument into Y. *Its the missing edges that matter for the arguments we will need to make.*



Some authors show unobserved common causes as a bidirected arc.

Terminology

1. nodes, vertices: the variables that are written
2. edges, arrows: the lines, may be “directed” or “bidirected” (arcs)
3. parents: for a given node, the causes feeding into it
4. root nodes: a node with no (endogenous) parents on the graph
5. descendants, children: effects of a given cause
6. paths: any series of connected nodes, regardless of direction
7. **d**irected **a**cyclic **g**raph: a graph where all edges are directed and you can never follow directed paths from a node back to itself.

Paths, dependencies, and information

The first use of graphs is to get a handle on what observables will be conditionally or marginally associated with what others.

Three key structures: forks, chains, colliders

Chain:

$$A \longrightarrow B \longrightarrow C$$

Implications:

$A \perp\!\!\!\perp C$?

$A \perp\!\!\!\perp C | B$?

$B \perp\!\!\!\perp C | A$?

Paths, dependencies, and information

Fork (aka common cause):

$$A \longleftarrow B \longrightarrow C$$

Implications:

$A \perp\!\!\!\perp C$?

$A \perp\!\!\!\perp C | B$?

$B \perp\!\!\!\perp C | A$?

Paths, dependencies, and information

Collider:

$$A \longrightarrow B \longleftarrow C$$

$A \perp\!\!\!\perp C$?

$A \perp\!\!\!\perp C | B$?

$B \perp\!\!\!\perp C | A$?

Beware colliders! Beware descendants of colliders!

Colliding is real

Pearl's favorite:

A: It rains

B: Sprinkler recently on

C: The lawn is wet

Hollywood success:

A. Good looks

B. Acting skills

C. Fame

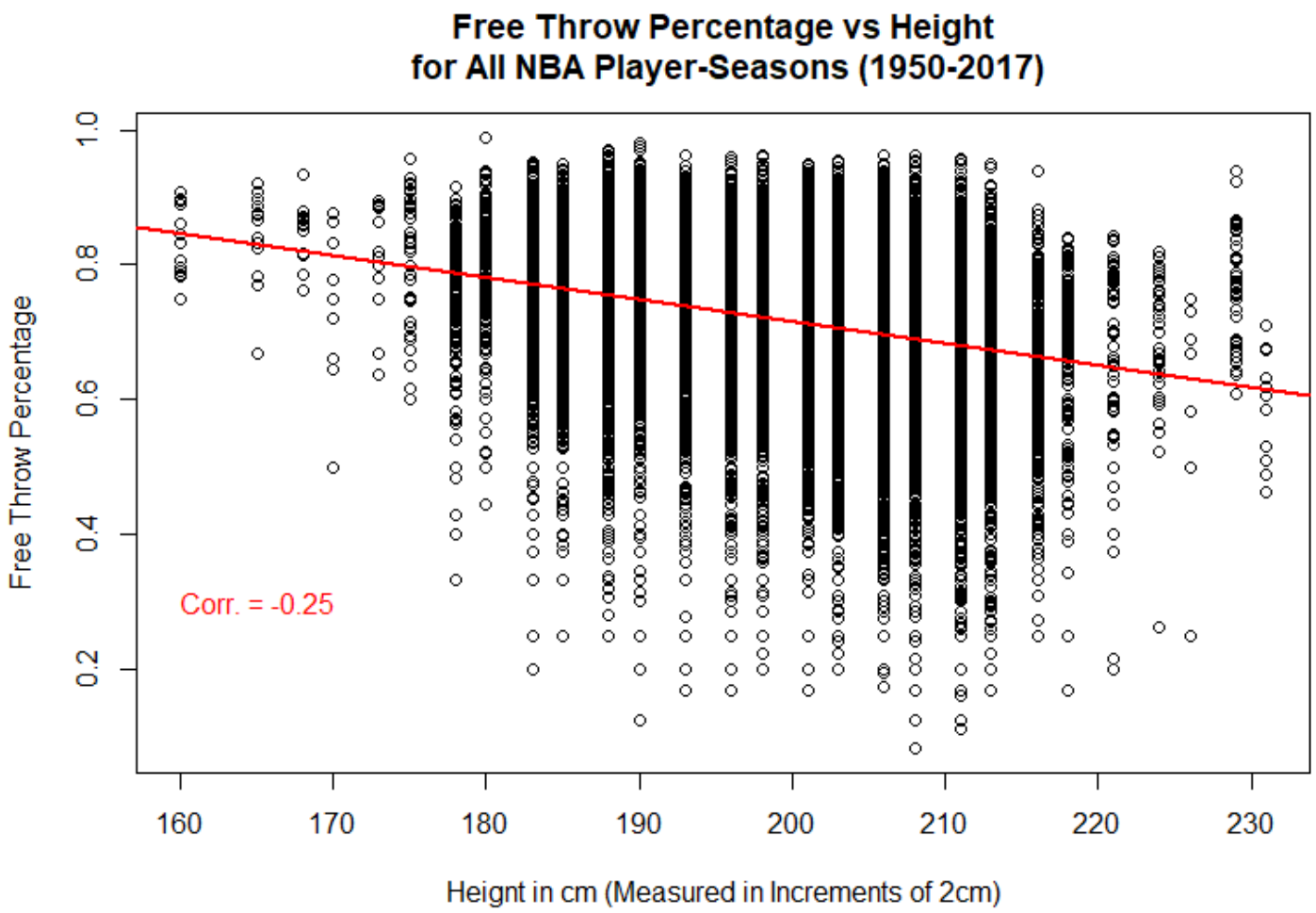
Academic tenure:

A: Productivity

B: Originality

C: Tenure[1]

[1] Thanks to Jason Roy for these examples



*Source: Adam Rohde, via `https://www.basketball-reference.com/` and `https://www.kaggle.com/datasets/drgilermo/nba-players-stats`.

d-Separation

You now understand d (directional)-Separation, which just follows from these to formalize when we can claim conditional independence

Rule

d-separation.
A and C are d-separated by a set of nodes Z if and only if:

- *A and C are connected by a chain ($A \rightarrow B \rightarrow C$) or fork ($A \leftarrow B \rightarrow C$) s.t. $B \in Z$, or*
- *A and C are connected by a collider ($A \rightarrow B \leftarrow C$) s.t. the collision node is not in Z, and no descendant of it is in Z.*

If nodes A and C are d-separated by Z, we expect $A \perp\!\!\!\perp B \mid Z$ in any dataset generated by that graph. The opposite is usually, but not always, true.

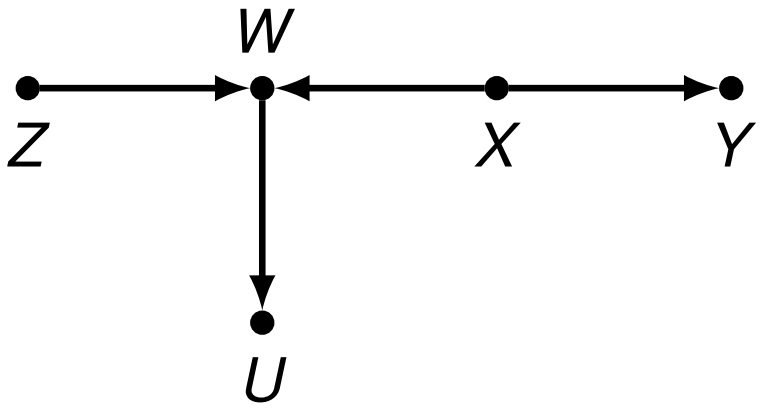
Warning: words that sound alike

Just in case you did not notice, conditional independence is not to be confused with conditional ignorability

- Conditional independence is about observables, e.g. $D \perp\!\!\!\perp Y | X$
- Conditional ignorability is about potential outcomes, e.g. $D \perp\!\!\!\perp Y_d | X$.

We will soon use conditional independence on DAGs to support claims of conditional ignorability, but not yet.

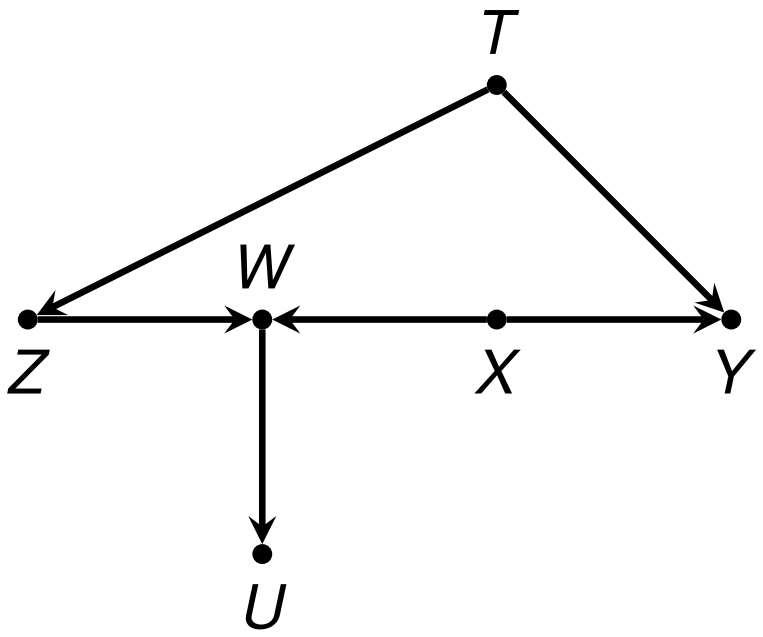
Practice



Looking at Z and Y:

- Are they d-separated conditional on $\{\}$?
- Are they d-separated conditional on W?
- Are they d-separated conditional on $\{W, X\}$?
- Are they d-separated conditional on U?

More practice



What sets d-separate Z from Y?

Review so far:

- Two variables come to be associated due to
  - one causing another (directly or through a chain)
  - a common cause
  - conditioning on a common consequence (collider)
- You can render them independent (*d*-separated) if,
  - they are independent already
  - you condition on common cause or intermediate variable
  - you don't condition on collider (or descendant thereof)

Frontdoor paths and backdoor paths

- A frontdoor path from D to Y begins with an arrow emanating out of D.
  - These carry effects of treatment, so usually we do not want to block these.
- Backdoor paths between D and Y point into D.
  - These are typically unwanted sources of association we will want to remove from what we measure.

Frontdoor paths and backdoor paths

The **backdoor criterion**:

- choose variables $\{Z\}$ that can block all backdoor paths from treatment to outcome, without including any descendants of treatment.
- (equivalent "trick": remove arrows out of D on your graph, and find $\{Z\}$ that can d-separate D and Y.)
- if you can do this, you can identify the effect of D on Y by conditioning on those $\{Z\}$.
- it also means you obtained $Y_d \perp\!\!\!\perp D|\{Z\}$, i.e. conditional ignorability, SOO. (more in a moment)
- in more natural terms: if you rule out "spurious" reasons for such an association, then you have only the causal reason left.

This is 93+% of what people use DAGs for.

What next?

Once you find a conditioning set Z that satisfies the backdoor criterion relative to D on Y, you can

- In DAG land: compute $p(Y|do(d))$ using the adjustment formula,

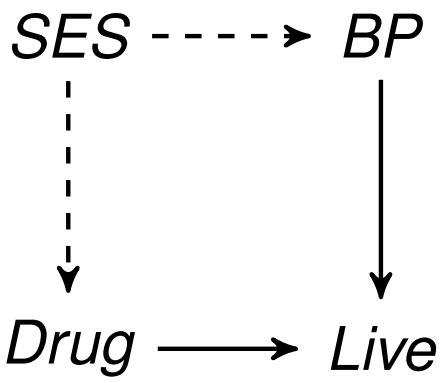
$$p(Y|do(d)) = \sum_z p(Y|d,z)p(z)$$

  - This is very general; you are more used to asking for the ATE, which you can get from this:

$$\begin{aligned}\mathbb{E}[Y_1 - Y_0] &= \sum_z y * p(Y = y|d = 1, z)p(z) - \sum_z y * p(Y = y|d = 0, z \\ &= \sum_z (\mathbb{E}[Y|d = 1, z] - \mathbb{E}[Y|d = 0, z])p(z) \\ &= \text{our usual approach under SOO}\end{aligned}$$

- In PO land: "you have shown SOO/CI, so condition on Z", i.e. estimate treatment effect within strata of Z, average over strata.
- In practice: use your favorite technology to "adjust for" Z.

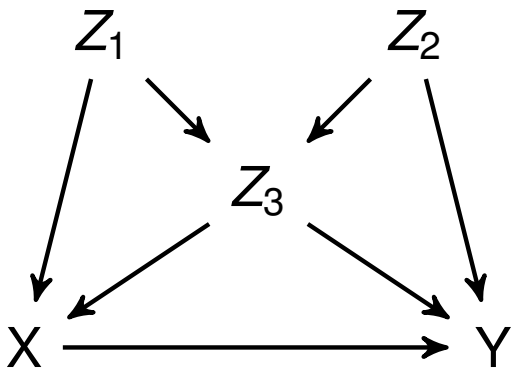
Example 1



where *SES* is socioeconomic status (unobserved), *BP* is blood pressure, *Drug* is a treatment, and *Live* is the outcome.

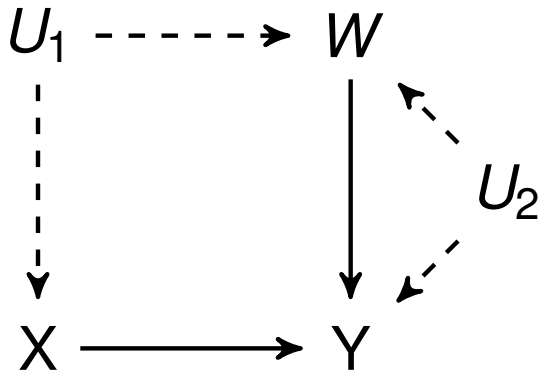
- Go for backdoor criterion: what do you choose?
- Condition on it.

Example 2



- What are the backdoor paths from X to Y?
- What sets block them?

How about here?



Adjustment Criterion

Shpitser, VanderWeele, Robins (2012)

Noting that $G_{\overline{X}}$ is the graph representing $do(X = x)$[2]

Rule

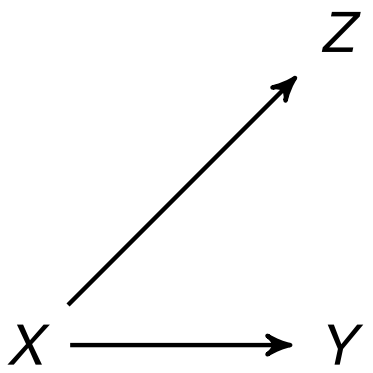
Adjustment criterion. *Z satisfies the adjustment criterion relative to (X, Y) in G if (i) No element in Z is a descendant in $G_{\overline{X}}$ of any $W \notin X$ which lies on a proper causal path from X to Y; (ii) All non-causal paths in G from X to Y are blocked by Z.*

Basically backdoor criterion, but allows conditioning on descendants of X when it is not on path to Y.

[2] Also relevant: this paper defines descendants of X to include X itself.

Adjustment criterion vs. Backdoor

Example:



Conditioning on Z fails backdoor for effect of X on Y, but not adjustment criterion.

- If it satisfies the backdoor criterion, it satisfies the adjustment criterion.
- Further, this is complete in that if Z fails to satisfy the adjustment criterion, then adjusting for Z will fail to produce a valid effect estimate.
- I don't know when this is useful in practice, but it is complete.
- Conditioning on such a Z could also weaken estimation, since conditionally on Z, variation in X would be lessened.

DAGs, POs, and conditional ignorability

Again,

If some $\{Z\}$ satisfies backdoor/adjustment criterion for Y-D, then $Y_d \perp\!\!\!\perp D \mid \{Z\}$ (conditional ignorability)

We won't show this rigorously, but try this on:

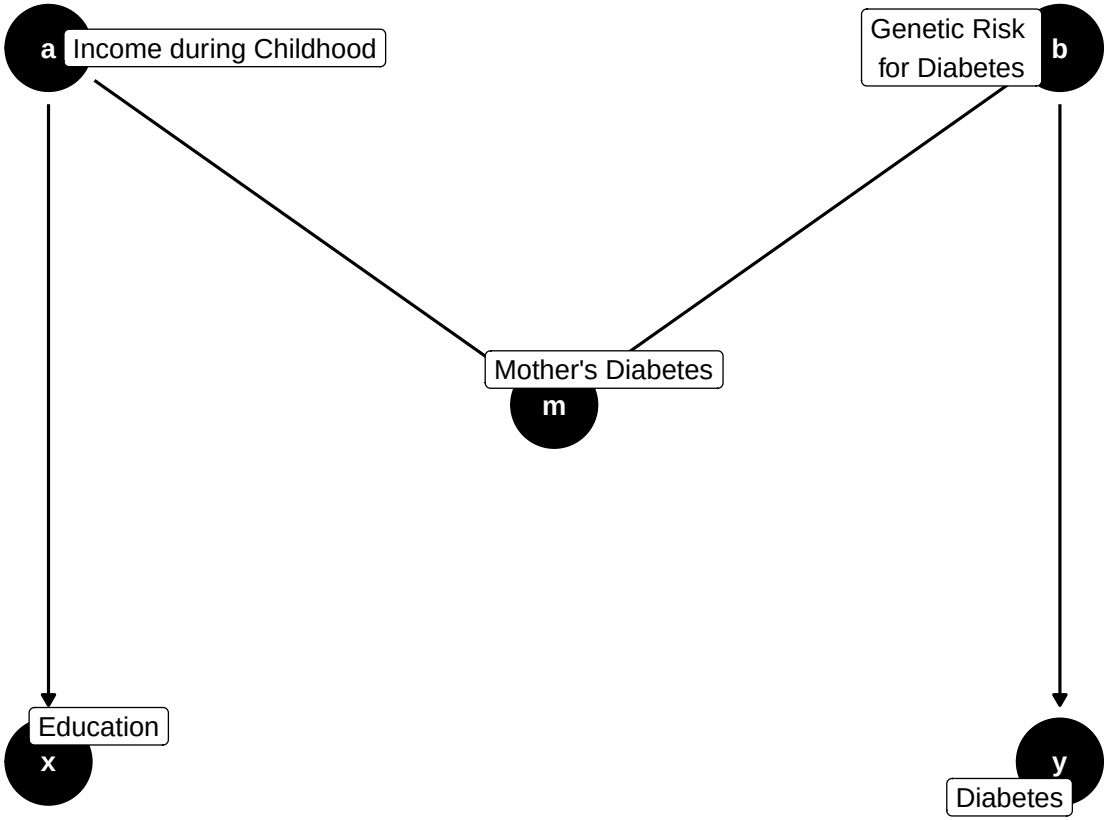
- "Satisfying backdoor means that, conditional on $\{Z\}$, there is no reason Y and D are related but for the effect of D on Y."
- "If, within strata of $\{Z\}$, you thought the only way D and Y were related was through the effect of treatment (i.e. backdoor holds), then within strata of $\{Z\}$, you'd have Y_0 independent of D. To help, you can imagine that looking at Y_0 "removes from Y the effect of D".
- Mostly people use DAGs just to check if conditioning on some $\{Z\}$ is a good idea, or to choose the $\{Z\}$ to condition on.
- DAGs and POs are compatible; DAGs help find failures in conditional ignorability claims. They are a thinking aid.

Why bother? Some errors and revelations made clear by DAGs

- M-bias, butterfly bias, sample selection
- Stratifying on post-treatment variables
- Why you cannot control for Z and check D-Y for exclusion restriction.

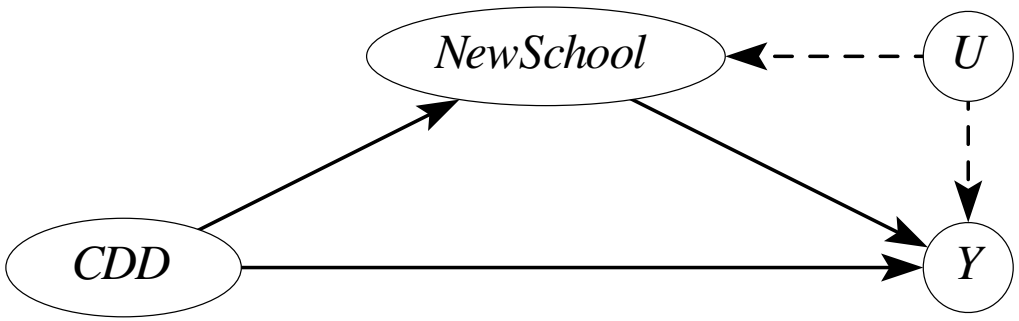
M-bias, butterly bias, sample selection

You already saw this but let's do another example[3]

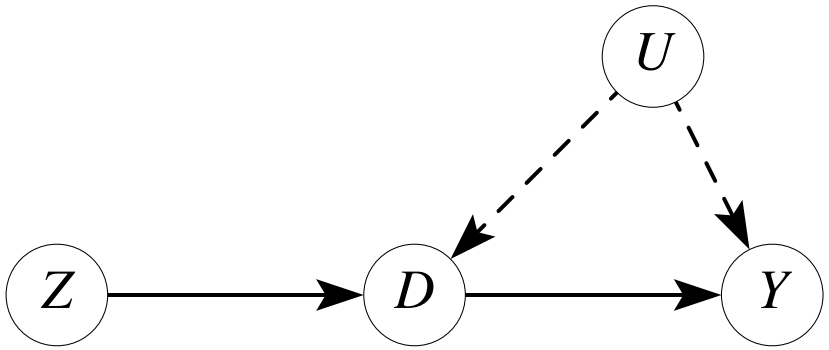


[3] Example and ggdag code courtesy of Malcolm Barrett, using an example borrowed from "Modern Epidemiology"

Stratifying on post-treatment trouble



You can't show exclusion by checking $Y \sim Z|D$



1. I can't hope to know the whole causal structure, so I prefer to just focus on POs.

Sorry, life really is that hard. The PO-based claims you'd like to make (e.g. CI/SOO) can be wrong due to the causal structure, so you can't wish it away. e.g.:

- If you can think of a plausible DAG that would not satisfy backdoor, there is a problem your CI story.
- Given colliders and such, this enables you to find challenges to CI that are not obvious

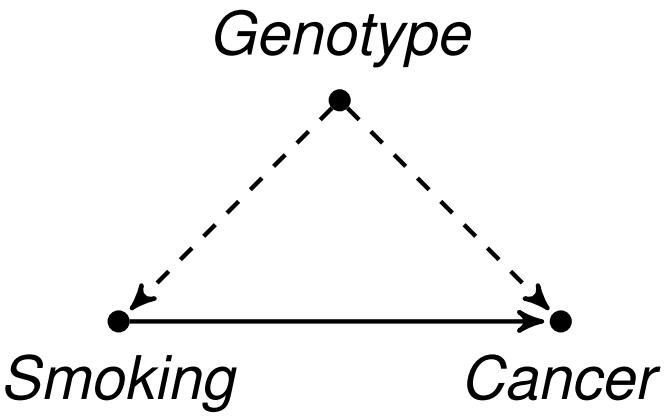
2. Many ID strategies (IV, RDD, DID) depend on assumptions not on a DAG, and therefore cannot be "proven" or fully understood from DAG perspective.

- Good reason to be bilingual and not *just* rely on DAGs.
- But these approaches still require assumptions that can be shown on DAG *plus* some others, so you can still surface problems from the DAG side
- In fact DAGs are great for explaining some things, e.g. IV and the exclusion restriction, even if that's not enough.

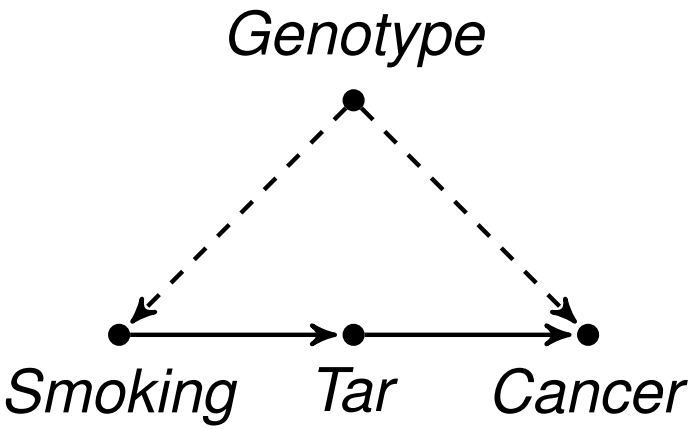
Front-door criterion

- Backdoor criterion is great, but doesn't capture all cases where non-parametric identification is possible.
- The best known extra case is the front-door.

Classic case: smoking and lung cancer

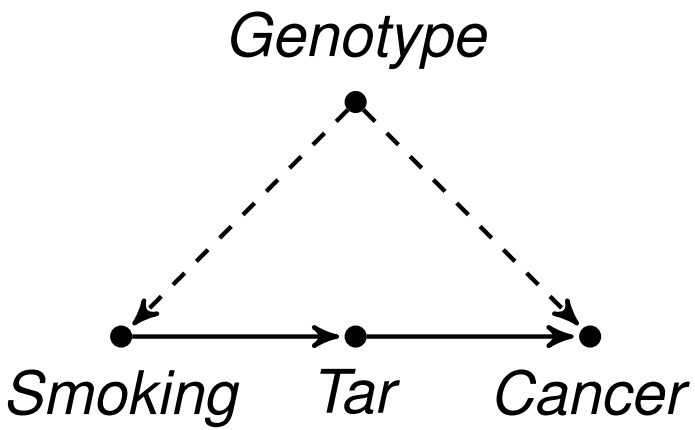


Hopeless



Solved

Shut the front door!



Can you ID the effect of smoking on tar? How?

$$p(t|do(s)) = p(t|s)$$

Can you ID the effect of tar on cancer? How?

$$p(c|do(t)) = \sum_s p(c|t,s)p(s)$$

Putting the pieces together

We have:

- When tar is set to level t, we see cancer $p(c|do(t))$ of the time.
- Smoking causes tar to be at that level as often as often as $p(t|do(s))$
- We can combine those into:

$$p(c|do(s)) = \sum_t p(c|do(t))p(t|do(s))$$

- And we have those pieces:

$$p(c|do(s)) = \sum_t \sum_{s'} p(C = c|T = t, S = s')p(S = s')p(T = t|S = s)$$

where s' is just an index, to be distinguished from s.

Are those all the ID strategies DAGs show us?

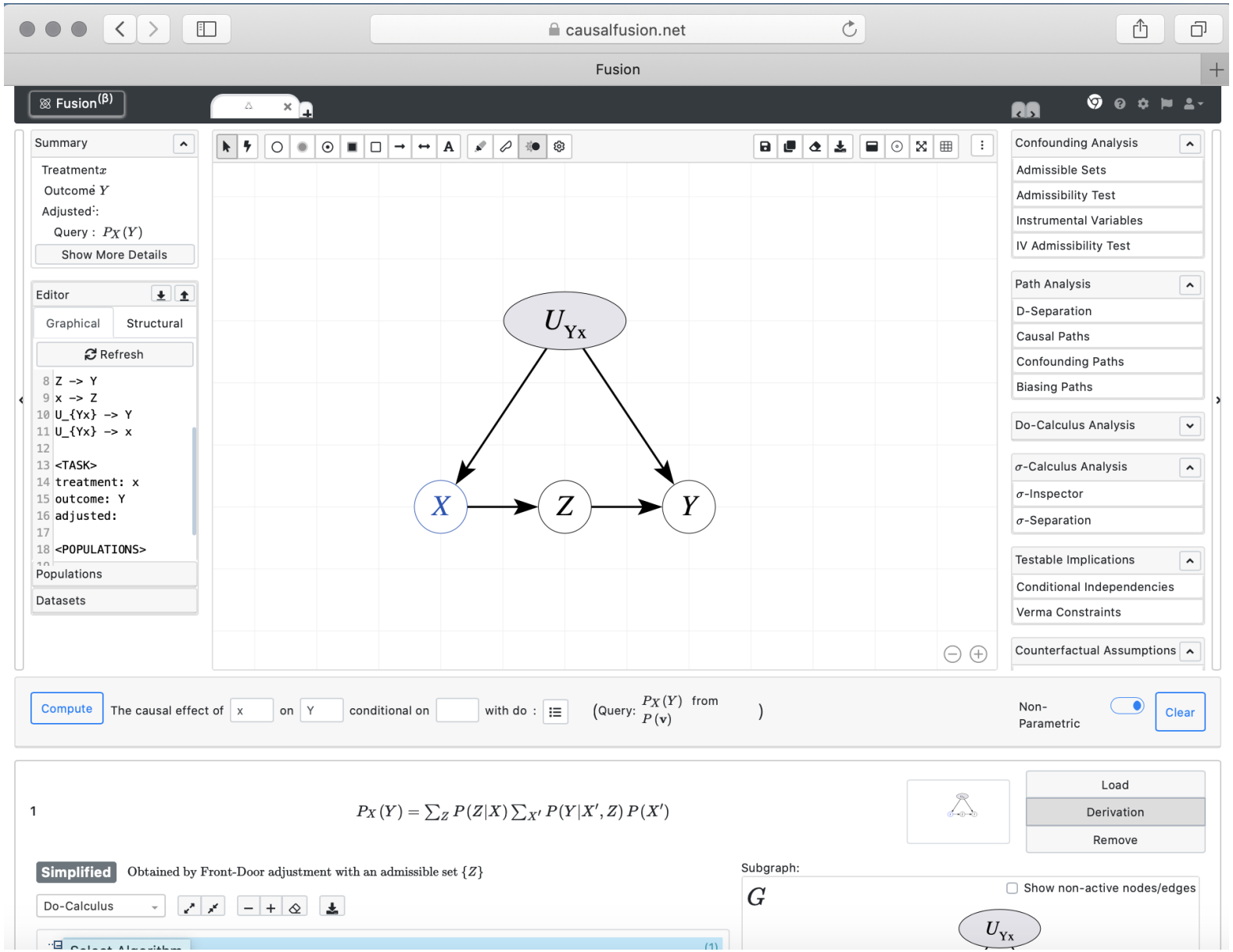
No. Backdoor and front-door criteria are great tricks, but don't cover every ID strategy justifiable by your graph.

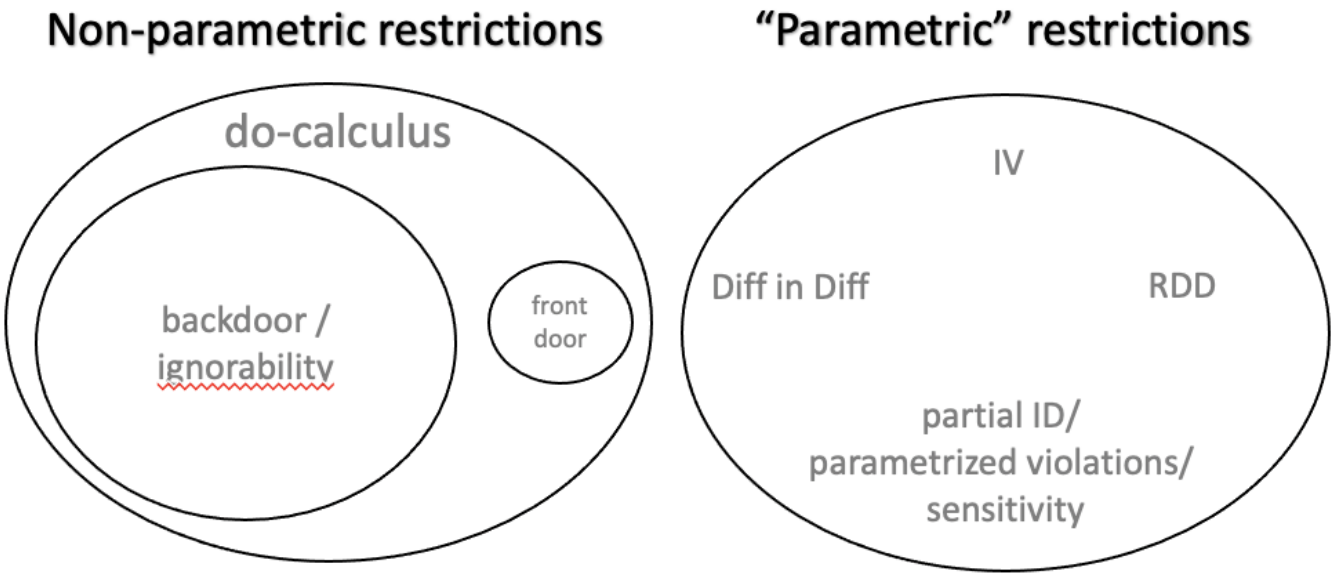
A *complete* guide is offered by the do-calculus

- Three equalities that you can use (with probability laws) to build any identification result that exists.
- Mostly, you know them already, we just need to write them a bit more comprehensively.
- It can still take work and creativity to figure out how to apply these rules to get a result, but there is also a algorithm that can find all results

We won't cover this, but:

Causal Fusion – give it a try



The identification landscape

Useful to have this picture in mind: